Term Project Proposal

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1. Business Question:

How does the level of homelessness in downtown Toronto affect local retail sales and foot traffic patterns?

2. Datasets to be Used:

We will use a dataset from the city of Toronto to answer our business question. Here is a link to the dataset: https://open.toronto.ca/dataset/daily-shelter-overnight-service-occupancy-capacity/

3. High-Level Description of the Dataset:

Our dataset includes 48794 observations and 32 variables. The most important variables that we believe can help us with our goal are: location, capacity, and service user count.

4. Similar Projects:

There are two examples that are similar to ours. The first one is called the Shelter System Flow, published by the City of Toronto, and the second one is the Downtown Recovery Dashboard, published by the University of Toronto.

Shelter System - City of Toronto:

The flow data of the Shelter System, developed by the City of Toronto, includes monthly information about how people enter, exit, and stay in the city's shelter system. The data indicated measures "active homelessness", which refers to people who have used a shelter in the previous three months and have not exited to permanent housing in that time. There are some basic metrics, like the inflow (newly identified, returns from permanent housing, or from other sheltered/homeless situations to the system), the outflow (to permanent housing or to becoming inactive), and the active homeless population itself. The monthly published data is disaggregated by subpopulations, such as families, single adults, youth, and the chronically homeless, where available. While the nightly occupancy datasets provide data on monthly shelter use, the Shelter System Flow data provides a much broader and longer view of trends in homelessness over time.

However, the Shelter System does have its limitations. It only includes monthly totals, which means you lose detailed views over days when analyzing short-term retail or foot traffic trends. Additionally, the data is for the city as a whole, and does not allow for the level of precision at the neighborhood level that your project will achieve. Furthermore, it only speaks to homelessness, without any link to economic indicators. Our project will combine daily shelter data (occupancy) with retail and pedestrian data (and control for weather, events, etc.), in order to be able to find more localized and predictive relationships between homelessness and downtown activity.

<u>Downtown Recovery Dashboard - University of Toronto:</u>

The Toronto Downtown Recovery Project, undertaken by the University of Toronto's School of Cities, evaluates downtown vitality through anonymized mobile data and tracks the number of unique visitors in downtowns across different cities. This research assesses downtown recovery after the pandemic by providing rankings, trends, and comparisons across cities. While this research provides a useful big-picture overview, it remains broad in perspective across multiple cities. In contrast to the Toronto Downtown Recovery Project, our project will be focusing on downtown Toronto and linking social factors like homelessness to economic outcomes in a more granular, local way. In our project, we will link the number of unhoused individuals with retail activity and foot traffic. By using daily shelter data and focusing on specific neighborhoods, we can get a clearer picture of how homelessness affects local economies.

References:

Toronto, C. O. (2024, January 22). *Shelter system flow data*. City of Toronto. https://www.toronto.ca/city-government/data-research-maps/research-reports/housing-and-homelessness-research-and-reports/shelter-system-flow-data/

Allen, (n.d.). *Downtown recovery*. School of Cities. https://downtownrecovery.com/